NICOLAS CEZAR DE CARVALHO BERREDO

Seattle, WA | nicodpf@uw.edu | 206-470-9764 | linkedin.com/in/nicolasberredo | nicolasberredo.com

EDUCATION

B.S. Electrical and Computer Engineering | University of Washington**,** Seattle, WA

Expected Graduation: June 2028 *GPA: 3.52*

Relevant Coursework: Signals & Systems, Electromagnetism, Data Structures & Algorithms, Intermediate Data Programming, Digital Circuits, Exploring Hardware Devices

Dean's List: Fall 2024, Spring 2025, Fall 2025

ENGINEERING EXPERIENCE

Undergraduate Research Assistant | UW ARC Lab - Project SHIELD | Seattle, WA *June 2026 – Present*

- Built an end-to-end Python pipeline for sensor health monitoring, generating 150+ GB of synthetic fault data by injecting sensor-specific Gaussian noise across multiple sensor types.
- Implemented signal-processing and machine-learning pipelines using MODWT, time-frequency features, Random Forest, and XGBoost to model sensor degradation and assess hardware faults.
- Parallelized large-scale sensor-data processing, achieving a 3.1× speedup in runtime.

R&D Electrical Team Member | E-Truck Challenge at UW | Seattle, WA *January – June 2026*

- Designed CAN J1939 message definitions in CANdb++ for a radiator-mounted cooling fan.
- Prototyped control and safety circuits for Project RAIJIN's high-voltage system. Used comparators, regulators, transducers, and timers.
- Calibrated and validated the safety interlock circuit to meet its specified trip threshold through bench testing and hardware debugging with an oscilloscope, digital multimeter and power supply.

Electronics Subteam Member | Reptiles PUC-Rio Baja SAE Team | Rio de Janeiro, Brazil *March – August 2024*

- Contributed to the electronics of a BAJA SAE off-road motorsports team, helping win a regional championship.
- Prototyped an embedded vibration-measurement system integrating an accelerometer, MCU, and SD module via I2C and SPI to evaluate suspension configurations and assess driver comfort following ISO 2631-1.
- Identified a suspension configuration with 62% lower measured vertical acceleration than the original configuration.
- Used RapidHarness to document wiring harnesses and SolidWorks to design a vehicle stand.

WORK EXPERIENCE

Operations Software Summer Intern | BTG Pactual Asset Management | Rio de Janeiro, Brazil *July – September 2026*

- Developed a Python optimization and simulation framework for target-date retirement ETFs, using Monte Carlo simulations and ARIMA forecasts based on historical market data to project annual returns exceeding 20%.
- Built an interactive dashboard for portfolio analytics and scenario visualization to support new financial products.

AI Developer Summer Intern | Maximiza Tecnologias | Rio de Janeiro, Brazil *June – September 2025*

- Led development of a WhatsApp ordering chatbot powered by OpenAI's LLM, supporting more than 100 customer inquiries daily.
- Developed a Streamlit application with an AI assistant to manage orders, query data, and automate notifications.

LEADERSHIP ACTIVITIES

President | Brazilian Student Association at UW | Seattle, WA *June 2026 – Present*

Co-Founder & Director of Logistics *March 2025 – June 2026*

- Built Notion dashboards to coordinate event planning and team handoffs; formalized the association's constitution.
- Planned 10+ events with 50 attendees each and raised US$1,500 while supporting 50+ Brazilian students in Seattle.

SKILLS & DISTINCTIONS

Embedded & Digital: Embedded systems, FPGAs, MCUs; Python, Java, SystemVerilog; Quartus Prime, ModelSim; Linux.
Circuits & Testing: KiCad, LTspice; SPI, I2C, CAN; hardware testing, soldering; waveform generator, DMM, oscilloscope.

Languages: Portuguese (native), English (fluent), Spanish (intermediate).

TEDxYouth Speaker | TEDxYouth@ColegioSantoInacio, 2023